Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

Line When asteroids collide, some collisions cause an asteroid to spin faster; others slow it down. If asteroids are all monoliths—single rocks—undergoing random collisions, a graph of their rotation rates (5) should show a bell-shaped distribution with statistical "tails" of very fast and very slow rotators. If asteroids are rubble piles, however, the tail representing the very fast rotators would be missing, because any loose aggregate spinning faster than once every few (10) hours (depending on the asteroid's bulk density) would fly apart. Researchers have discovered that all but five observed asteroids obey a strict limit on rate of rotation. The exceptions are all smaller than 200 meters in diameter, with an abrupt cutoff for (15) asteroids larger than that.

The evident conclusion—that asteroids larger than 200 meters across are multicomponent structures or rubble piles—agrees with recent computer modeling of collisions, which also finds a transition at that (20) diameter. A collision can blast a large asteroid to bits, but after the collision, those bits will usually move slower than their mutual escape velocity. Over several hours, gravity will reassemble all but the fastest pieces into a rubble pile. Because collisions among (25) asteroids are relatively frequent, most large bodies have already suffered this fate. Conversely, most small asteroids should be monolithic, because impact fragments easily escape their feeble gravity.

Questions 606–609 refer to the passage.

606. The passage implies which of the following about the five asteroids mentioned in line 12?

A. Their rotation rates are approximately the same.

B. They have undergone approximately the same number of collisions.

C. They are monoliths.

D. They are composed of fragments that have escaped the gravity of larger asteroids.

E. They were detected only recently.

607. The discovery of which of the following would call into question the conclusion mentioned in line 16?

A. An asteroid 100 meters in diameter rotating at a rate of once per week

B. An asteroid 150 meters in diameter rotating at a rate of 20 times per hour

C. An asteroid 250 meters in diameter rotating at a rate of once per week

D. An asteroid 500 meters in diameter rotating at a rate of once per hour

E. An asteroid 1,000 meters in diameter rotating at a rate of once every 24 hours

608. According to the passage, which of the following is a prediction that is based on the strength of the gravitational attraction of small asteroids?

A. Small asteroids will be few in number.

B. Small asteroids will be monoliths.

C. Small asteroids will collide with other asteroids very rarely.

D. Most small asteroids will have very fast rotation rates.

E. Almost no small asteroids will have very slow rotation rates.

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

609. The author of the passage mentions "escape velocity" (see line 22) in order to help explain which of the following?

A. The tendency for asteroids to become smaller rather than larger over time

B. The speed with which impact fragments reassemble when they do not escape an asteroid's gravitational attraction after a collision

C. The frequency with which collisions among asteroids occur

D. The rotation rates of asteroids smaller than 200 meters in diameter

E. The tendency for large asteroids to persist after collisions

Line Most attempts by physicists to send particles faster than the speed of light involve a remarkable phenomenon called quantum tunneling, in which particles travel through solid barriers that appear (5) to be impenetrable. If you throw a ball at a wall, you expect it to bounce back, not to pass straight through it. Yet subatomic particles perform the equivalent feat. Quantum theory says that there is a distinct, albeit small, probability that such a particle (10) will tunnel its way through a barrier; the probability declines exponentially as the thickness of the barrier increases. Though the extreme rapidity of quantum tunneling was noted as early as 1932, not until 1955 was it hypothesized—by Wigner and (15) Eisenbud—that tunneling particles sometimes travel faster than light. Their grounds were calculations that suggested that the time it takes a particle to tunnel through a barrier increases with the thickness of the barrier until tunneling time (20) reaches a maximum; beyond that maximum, tunneling time stays the same regardless of barrier thickness. This would imply that once maximum tunneling time is reached, tunneling speed will increase without limit as barrier thickness increases. Several recent (25) experiments have supported this hypothesis that tunneling particles sometimes reach superluminal speed. According to measurements performed by Raymond Chiao and colleagues, for example, photons can pass through an optical filter at 1.7 times the (30) speed of light.

Questions 610–612 refer to the passage.

610. The author of the passage mentions calculations about tunneling time and barrier thickness in order to

A. suggest that tunneling time is unrelated to barrier thickness

B. explain the evidence by which Wigner and Eisenbud discovered the phenomenon of tunneling

C. describe data recently challenged by Raymond Chiao and colleagues

D. question why particles engaged in quantum tunneling rarely achieve extremely high speeds

E. explain the basis for Wigner and Eisenbud's hypothesis

611. The passage implies that if tunneling time reached no maximum in increasing with barrier thickness, then

A. tunneling speed would increase with barrier thickness

B. tunneling speed would decline with barrier thickness

C. tunneling speed would vary with barrier thickness

D. tunneling speed would not be expected to increase without limit

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

D. tunneling speed would not be expected to increase without limit

E. successful tunneling would occur even less frequently than it does

612. Which of the following statements about the earliest scientific investigators of quantum tunneling can be inferred from the passage?

A. They found it difficult to increase barrier thickness continually.

B. They anticipated the later results of Chiao and his colleagues.

C. They did not suppose that tunneling particles could travel faster than light.

D. They were unable to observe instances of successful tunneling.

E. They made use of photons to study the phenomenon of tunneling.

DO NOT COPY
filippo.fiz@uptoten.it